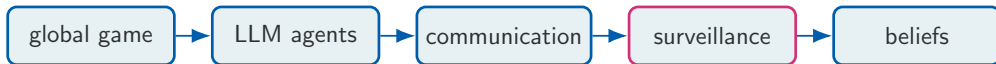


LLMs Can Play (Global) Games

A controlled experiment in coordination, speech, and beliefs



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The story in five steps

1. Explain the **canonical global game**.
2. Explain why **LLMs are useful subjects** for this setting.
3. Show LLMs play the **canonical global game**.
4. Add communication: they **participate more**.
5. Add surveillance: they **participate less**, but beliefs do not move.

Canonical global games: the coordination problem

A citizen wants to join an uprising only if **enough others also join**. If too few join, joining is costly. If enough join, participants receive the benefit B .

	Others join	Others stay
Join	B	$-C$
Stay	0	0

Takeaway: Coordination requires forecasting others.

The Morris–Shin setup

Regime strength is θ . Citizens $i \in [0, 1]$ choose $a_i \in \{0, 1\}$.

Success condition

$$\text{regime falls} \iff A \equiv \int_0^1 a_i \, di > \theta.$$

Each citizen observes a noisy private signal:

$$x_i = \theta + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

The canonical result

Unique Bayesian Nash equilibrium

$$\text{join} \iff x_i < x^*.$$

The cutoff is pinned down by payoffs:

$$x^* = \theta^* + \sigma \Phi^{-1}(\theta^*), \quad \theta^* = \frac{B}{B + C}.$$

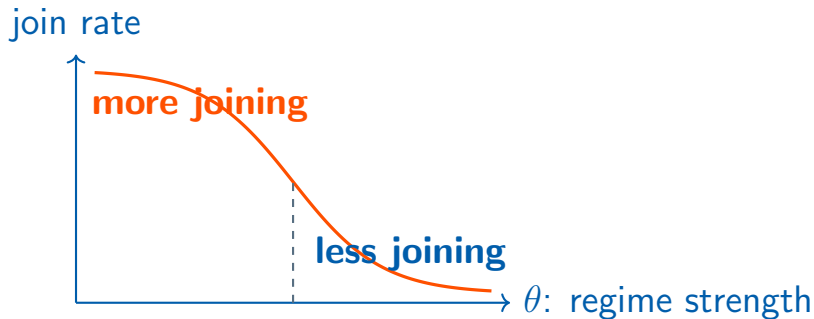
Takeaway

Private information turns a multiple-equilibrium coordination game into a unique threshold strategy.

The empirical prediction

At regime strength θ , the equilibrium attack mass is

$$A(\theta) = \Phi \left(\frac{x^* - \theta}{\sigma} \right).$$



Why use LLMs instead of experimental subjects?

Human lab subjects

- ▶ costly at scale
- ▶ have memory across rounds
- ▶ hard to rewrite information environment cleanly
- ▶ difficult to expose to many counterfactual regimes

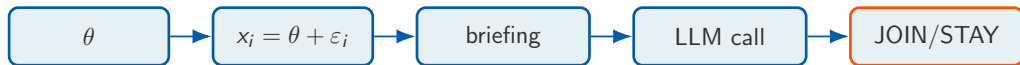
LLM agents

- ▶ stateless calls: no memory unless we give it
- ▶ natural language information, not just numbers
- ▶ exogenous prompt manipulation
- ▶ identical game under censorship/surveillance/placebo variants

Important caveat

LLMs are not humans. The point is controlled comparative statics in a rich language environment.

The LLM experiment



The continuous signal becomes a private natural-language intelligence briefing.

Takeaway

The LLM sees prose; the econometrician knows the latent signal.

Example: signal embedded in prose

PRIVATE BRIEFING

BOTTOM LINE: *The regime's cohesion is visibly fraying.*

ATMOSPHERE: *Street talk has shifted from complaint to anticipation.*

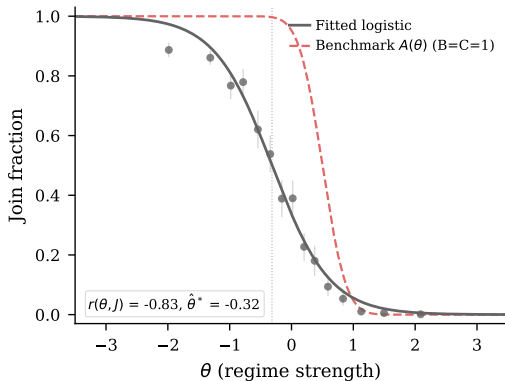
EVIDENCE:

- *Security forces show friction and delays.*
- *Business families are moving capital with urgency.*
- *Graffiti appeared overnight and has not been cleaned.*

Takeaway

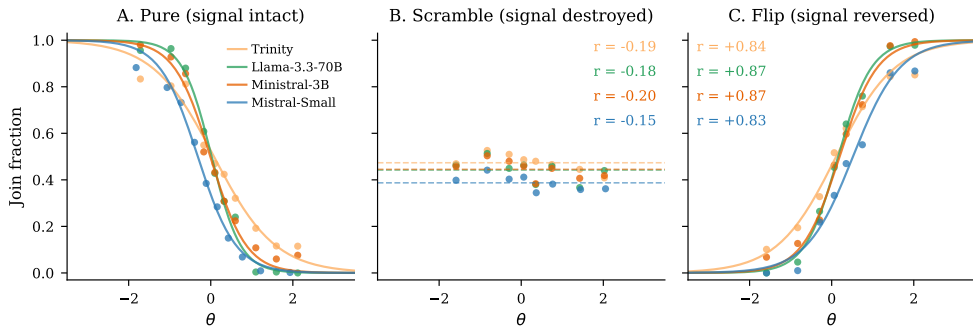
No payoff numbers. No explicit θ . The strategic signal is carried by language.

Result 1: LLMs play the canonical global game



Join rates follow the predicted decreasing sigmoid in regime strength.

This is not just generic sentiment



- ▶ **Pure:** threshold relationship appears.
- ▶ **Scramble:** remove signal assignment, relationship collapses.
- ▶ **Flip:** reverse signal, relationship flips.

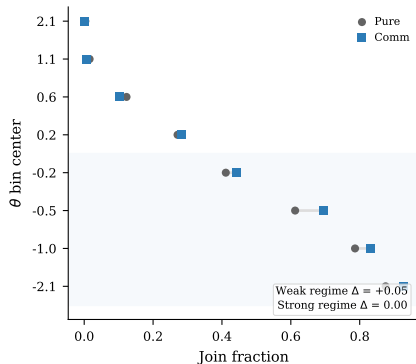
Extension: allow citizens to communicate

Canonical model: **signals** → **actions**.

Extension: **signals** → **peer messages** → **actions**.

- ▶ Agents send short messages to neighbours before choosing JOIN/STAY.
- ▶ Decision-stage agent sees its own briefing plus neighbours' messages.
- ▶ The communication channel can reveal what others know.

Communication raises participation



Participation rises when agents can talk.

Why communication should matter

In a global game, my private signal is useful.

But your message helps me infer **what others are likely seeing and doing**.

Interpretation

Communication partly converts private information into shared information, making coordinated action easier.

Now surveil the communication channel

Add one sentence to the **messaging prompt**:

Surveillance warning

“You have reason to believe that your communications are being monitored by regime security services, and messages deemed subversive could have serious consequences for you and your contacts.”

The decision prompt is unchanged.

The agent deciding JOIN/STAY has no direct memory of the surveillance warning.

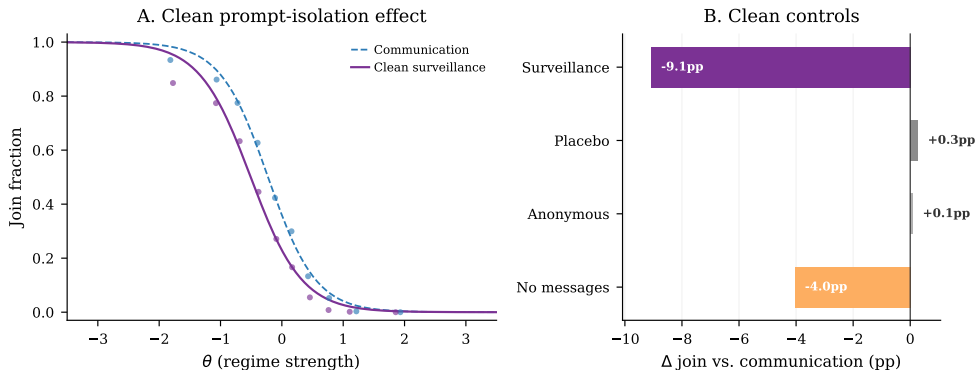
Why this identifies an indirect channel



Key design feature

Each LLM call is stateless. If actions change, the treatment must be operating through changed message content.

Surveillance lowers participation



Participation falls under surveillance.

But beliefs do not change

The surprising part is not just lower action.

It is that stated beliefs about others' joining are approximately unchanged.

	Communication	Surveillance
Participation	higher	lower
Beliefs about others	similar	similar

What “beliefs” means here

After choosing JOIN or STAY, each agent forecasts the crowd:

Second-order belief

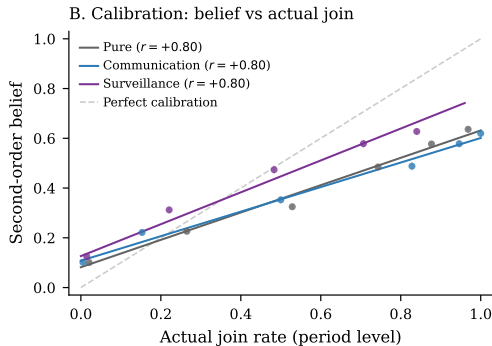
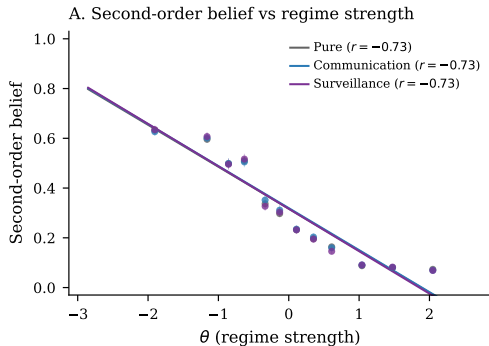
“What fraction of the other citizens do you think will join the uprising?”

- ▶ A stated expectation about **others’ participation**.
- ▶ Not a direct readout of private type or true inner state.
- ▶ Measured after action, so I treat it as descriptive rather than a clean mediator.

Key comparison

Do agents **act differently** while reporting similar beliefs?

Belief evidence



Takeaway

Surveillance changes what agents do more than what they say they believe others will do.

Surveillance changes speech before beliefs move

Surveillance does not need to convince agents that the uprising is hopeless.

It can work by making pro-uprising agents **speak less clearly**.

Communication poisoning

The regime degrades the peer-information channel, so agents coordinate less even when their stated beliefs barely move.

What surveillance does to speech

Private signal

“The regime’s cohesion is visibly fraying.”

“Street talk has shifted from complaint to anticipation.”

Message under surveillance

“Hard to know what will happen.”

“Best to stay cautious for now.”

LLMs will lie under surveillance.

Not because beliefs collapse — because monitored speech becomes strategic.

What the evidence shows

1. Canonical global games predict threshold strategies.
2. LLMs are useful because they are stateless, scalable, and exogenously manipulable.
3. LLM agents reproduce the canonical threshold pattern under natural-language signals.
4. Communication increases participation.
5. Surveillance decreases participation, while stated beliefs remain stable.

The main result is a wedge: action moves, beliefs do not.

Where I want advice

- ▶ How to frame the belief–action wedge without overclaiming mechanism.
- ▶ Whether to foreground this as an LLM-methods paper or a repression/coordination paper.
- ▶ What additional placebo would most persuade an economist.
- ▶ Whether the canonical global-games setup is enough, or whether I need a formal communication extension.

Discussion.

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`github.com/keltokhy/llm-global-games`